



# CUBAN NATIONAL CHEMISTRY CONTEST 2018

*“The future of our country must necessarily be a future of men of sciences, must be a future of men of thinking, because that is precisely what we are creating...”*

*Fidel Castro*

Name(s): \_\_\_\_\_ Last name(s): \_\_\_\_\_

School: \_\_\_\_\_

Municipality: \_\_\_\_\_ Province: \_\_\_\_\_

CODE: \_\_\_\_\_ Test: **DAY 2. TENTH GRADE EXAM**

## Instructions

- Write clearly all your data in the appropriate place (names, last names, school, municipality, province). Do not write anything in the field CODE.
- Verify that your questions booklet is complete. This is the Day 2 problem set, with an individual exam for 10<sup>th</sup> grade. The exam has 3 problems (problem 5, 7 items; problem 6, 6 items; problem 7, 3 items) and a total of 5 pages.
- All data, constants, conversions and basic equations needed are included in the exam. You are allowed to use a scientific calculator.
- At the beginning of each problem its scoring (from a total of 100 points), each item scoring (in marks) is indicated.

Success!



Ministerio de Educación  
República de Cuba



# CONSTANTS, CONVERSIONS AND EQUATIONS

|                                                                                                                 |                                                                              |                                                                                   |                                                                                            |
|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Avogadro's Constant                                                                                             | $N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$                                | Ideal gas equation                                                                | $p \cdot V = n \cdot R \cdot T$                                                            |
| Gasses' constant                                                                                                | $R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$             | Thermodynamics First Principle                                                    | $Q = \Delta E + W$                                                                         |
| Cero in the Celsius scale                                                                                       | 273,15 K                                                                     | Gibbs' energy                                                                     | $\Delta G = \Delta H - T \cdot \Delta S$                                                   |
| Faraday's constant                                                                                              | $F = 96485 \text{ C} \cdot \text{mol}^{-1}$                                  | Nernst's equation                                                                 | $E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \frac{c_{\text{red}}}{c_{\text{ox}}}$ |
| Water ionic product at 25 °C                                                                                    | $K_W = 1,00 \cdot 10^{-14}$                                                  | Arrhenius' equation                                                               | $k = A \cdot \exp \left( -\frac{E_A}{R \cdot T} \right)$                                   |
| Standard conditions                                                                                             | $p^\circ = 100 \text{ kPa}$<br>$c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$ | $\Delta G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$ |                                                                                            |
| $1 \text{ J} = 1 \text{ C} \cdot 1 \text{ V} = 1 \text{ kPa} \cdot 1 \text{ L} = 1 \text{ W} \cdot 1 \text{ s}$ |                                                                              | $1 \text{ C} = 1 \text{ A} \cdot 1 \text{ s}$                                     |                                                                                            |

## PERIODIC TABLE OF THE ELEMENTS

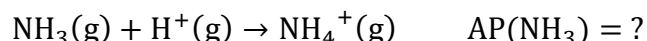
| PERIODIC TABLE OF THE ELEMENTS |                   |                   |                    |                    |                    |                    |                    |                    |                    |                    |                     |                    |                     |                    |                     |                   |                     |          |
|--------------------------------|-------------------|-------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|---------------------|--------------------|---------------------|--------------------|---------------------|-------------------|---------------------|----------|
| 1<br>1A                        |                   |                   |                    |                    |                    |                    |                    |                    |                    |                    |                     | 13<br>3A           |                     | 14<br>4A           | 15<br>5A            | 16<br>6A          | 17<br>7A            | 18<br>8A |
| 1<br>H<br>1.008                | 2<br>2A           |                   |                    |                    |                    |                    |                    |                    |                    |                    |                     | 5<br>B<br>10.81    | 6<br>C<br>12.01     | 7<br>N<br>14.01    | 8<br>O<br>16.00     | 9<br>F<br>19.00   | 10<br>Ne<br>20.18   |          |
| 3<br>Li<br>6.941               | 4<br>Be<br>9.012  |                   |                    |                    |                    |                    |                    |                    |                    |                    |                     | 13<br>Al<br>26.98  | 14<br>Si<br>28.09   | 15<br>P<br>30.97   | 16<br>S<br>32.07    | 17<br>Cl<br>35.45 | 18<br>Ar<br>39.95   |          |
| 11<br>Na<br>22.99              | 12<br>Mg<br>24.31 | 3<br>3B           | 4<br>4B            | 5<br>5B            | 6<br>6B            | 7<br>7B            | 8<br>8B            | 9<br>8B            | 10<br>8B           | 11<br>1B           | 12<br>2B            | 31<br>Ga<br>69.72  | 32<br>Ge<br>72.61   | 33<br>As<br>74.92  | 34<br>Se<br>78.96   | 35<br>Br<br>79.90 | 36<br>Kr<br>83.80   |          |
| 19<br>K<br>39.10               | 20<br>Ca<br>40.08 | 21<br>Sc<br>44.96 | 22<br>Ti<br>47.88  | 23<br>V<br>50.94   | 24<br>Cr<br>52.00  | 25<br>Mn<br>54.94  | 26<br>Fe<br>55.85  | 27<br>Co<br>58.93  | 28<br>Ni<br>58.69  | 29<br>Cu<br>63.55  | 30<br>Zn<br>65.39   | 49<br>In<br>114.8  | 50<br>Sn<br>118.7   | 51<br>Sb<br>121.8  | 52<br>Te<br>127.6   | 53<br>I<br>126.9  | 54<br>Xe<br>131.3   |          |
| 37<br>Rb<br>85.47              | 38<br>Sr<br>87.62 | 39<br>Y<br>88.91  | 40<br>Zr<br>91.22  | 41<br>Nb<br>92.91  | 42<br>Mo<br>95.94  | 43<br>Tc<br>(98)   | 44<br>Ru<br>101.1  | 45<br>Rh<br>102.9  | 46<br>Pd<br>106.4  | 47<br>Ag<br>107.9  | 48<br>Cd<br>112.4   | 81<br>Tl<br>204.4  | 82<br>Pb<br>207.2   | 83<br>Bi<br>209.0  | 84<br>Po<br>(209)   | 85<br>At<br>(210) | 86<br>Rn<br>(222)   |          |
| 55<br>Cs<br>132.9              | 56<br>Ba<br>137.3 | 57<br>La<br>138.9 | 72<br>Hf<br>178.5  | 73<br>Ta<br>180.9  | 74<br>W<br>183.8   | 75<br>Re<br>186.2  | 76<br>Os<br>190.2  | 77<br>Ir<br>192.2  | 78<br>Pt<br>195.1  | 79<br>Au<br>197.0  | 80<br>Hg<br>200.6   |                    | 114<br>Uuq<br>(277) |                    | 116<br>Uuh<br>(277) |                   | 118<br>Uuo<br>(277) |          |
| 87<br>Fr<br>(223)              | 88<br>Ra<br>(226) | 89<br>Ac<br>(227) | 104<br>Rf<br>(261) | 105<br>Db<br>(262) | 106<br>Sg<br>(263) | 107<br>Bh<br>(262) | 108<br>Hs<br>(265) | 109<br>Mt<br>(266) | 110<br>Ds<br>(269) | 111<br>Rg<br>(272) | 112<br>Uub<br>(277) |                    |                     |                    |                     |                   |                     |          |
|                                |                   |                   |                    |                    |                    |                    |                    |                    |                    |                    |                     |                    |                     |                    |                     |                   |                     |          |
|                                |                   | 58<br>Ce<br>140.1 | 59<br>Pr<br>140.9  | 60<br>Nd<br>144.2  | 61<br>Pm<br>(145)  | 62<br>Sm<br>150.4  | 63<br>Eu<br>152.0  | 64<br>Gd<br>157.3  | 65<br>Tb<br>158.9  | 66<br>Dy<br>162.5  | 67<br>Ho<br>164.9   | 68<br>Er<br>167.3  | 69<br>Tm<br>168.9   | 70<br>Yb<br>173.0  | 71<br>Lu<br>175.0   |                   |                     |          |
|                                |                   | 90<br>Th<br>232.0 | 91<br>Pa<br>231.0  | 92<br>U<br>238.0   | 93<br>Np<br>(237)  | 94<br>Pu<br>(244)  | 95<br>Am<br>(243)  | 96<br>Cm<br>(247)  | 97<br>Bk<br>(247)  | 98<br>Cf<br>(251)  | 99<br>Es<br>(252)   | 100<br>Fm<br>(257) | 101<br>Md<br>(258)  | 102<br>No<br>(259) | 103<br>Lr<br>(262)  |                   |                     |          |

**Problem 5. Thermochemistry (20 points)**

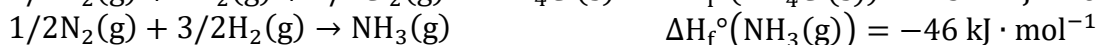
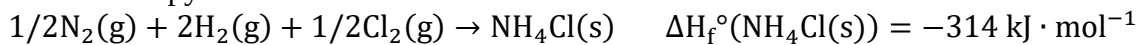
| 5.1 | 5.2 | 5.3 | 5.4 | 5.5 | 5.6 | 5.7 | Total | Total |
|-----|-----|-----|-----|-----|-----|-----|-------|-------|
| 10  | 3   | 4   | 8   | 3   | 4   | 8   | 40    | 20    |
|     |     |     |     |     |     |     |       |       |

Proton affinity (AP) is the energy associated with the process in which a chemical species gains a proton, an  $H^+$  ion. The value of this physical quantity provides quantitative information about the acid-base characteristics.

**5.1.** Use the data given to calculate the proton affinity of ammonia ( $NH_3$ ).



Standard enthalpy of formation:



Standard enthalpy of dissociation:



Electron affinity:



Ionization energy:



Lattice energy:



**5.2.** A substance is a good base when it can easily accept a proton (ion  $H^+$ ). Proton affinities of phosphine ( $PH_3$ ) and ammonia are  $-785 \text{ kJ} \cdot \text{mol}^{-1}$  and  $-854 \text{ kJ} \cdot \text{mol}^{-1}$ , respectively. Which of the two is the best base?

**5.3.** Phosphine and ammonia molecules are isoelectronic, but HPH and HNH bond angles have an amplitude of  $93,5^\circ$  and  $107,8^\circ$ , respectively. Which atomic orbitals are involved to a more significant extent in the formation of P-H and N-H bonds?

**5.4.** Represent the electronic configuration of ammonia and phosphine molecules according to the Valence Bond Theory. Only use the orbitals in the valence shell for your analysis.

The enthalpy of a covalent bond A-B, where A and B are two atoms, is the average energy needed to break one mole of these bonds at constant pressure. In diatomic molecules, the bond enthalpy is numerically equal to the dissociation enthalpy.

**5.5.** The standard mean bond enthalpies for P-H and N-H bonds are  $322$  and  $388 \text{ kJ} \cdot \text{mol}^{-1}$ , respectively. Justify why the N-H bond is stronger.

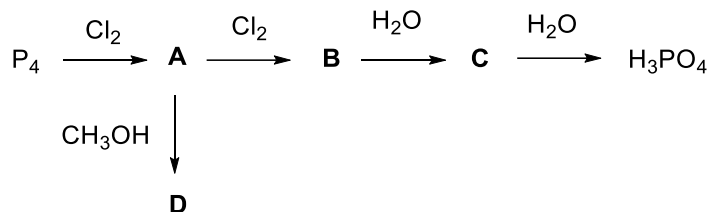
**5.6.** The strongest chemical bond known to date is that of the dinitrogen molecule. Explain this experimental fact.

**5.7.** Calculate the standard enthalpy of dissociation of the dinitrogen molecule. Elaborate on an appropriate thermodynamic cycle and use the data from the problem.

**Problem 6. Chemical bonding (15 points)**

| 6.1 | 6.2 | 6.3 | 6.4 | 6.5 | 6.6 | Total | Total |
|-----|-----|-----|-----|-----|-----|-------|-------|
| 8   | 3   | 2   | 2   | 9   | 6   | 30    | 15    |
|     |     |     |     |     |     |       |       |

The following diagram contains some of the reactions of tetraphosphorous and its compounds.



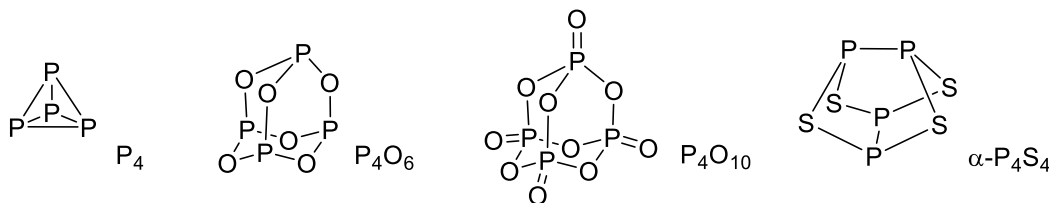
**6.1.** Find the substances represented by letters **A**, **B**, **C** and **D**.

**6.2.** Though the most stable allotropic modification of this element is  $\text{P}_4$  (in the form of white or red phosphorous), it is possible to detect a diatomic molecule in the gaseous phase. Predict its magnetic properties by using the Molecular Orbital Theory.

**6.3.** Phosphorus(V) oxide,  $\text{P}_4\text{O}_{10}$ , is a strong dehydrating reagent that reacts with nitric acid to form metaphosphoric acid,  $\text{HPO}_3$  and a white ionic solid. Write down the equation corresponding to this chemical reaction.

**6.4.** Phosphine,  $\text{PH}_3$ , is a highly toxic compound; it can be obtained by disproportionation tetraphosphorous in the presence of potassium hydroxide. Write the chemical equation knowing that potassium dihydrogen hypophosphate,  $\text{KH}_2\text{PO}_2$ , is also formed.

Phosphorous element is known for its capacity to form molecules and inorganic ions with great molecular complexity when combined with non-metallic elements. In these substances, it is usual to find cycles with various sizes in which the phosphorous atoms alternate with atoms of another chemical element.



**6.5.** Draw Lewis structures for the species diphosphate,  $\text{P}_2\text{O}_7^{4-}$ ; hypophosphate,  $\text{P}_2\text{O}_6^{4-}$ ; and hypophosphorous acid,  $\text{H}_3\text{PO}_2$ .

**6.6.** Suggest Lewis structures for the species  $\text{P}_3\text{O}_9^{3-}$  and  $(\text{NPCl}_2)_3$  if both molecules are symmetric and contain a six-atom cycle.

**Problem 7. Stoichiometry II (15 points)**

| 7.1 | 7.2 | 7.3 | Total | Total |
|-----|-----|-----|-------|-------|
| 10  | 10  | 10  | 30    | 15    |
|     |     |     |       |       |

It is a common practice to express the formula of a mineral as a mixture of oxides. For example, the formula of kaolin, the main constituent of clay, can be described as  $\text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2 \cdot 2\text{H}_2\text{O}$ . In this problem, your task will be to find the formula (oxide composition) of several important minerals.

Beryl is the most important mineral of beryllium and is a ternary oxide of beryllium, silicon and aluminum. In beryl Be:Al mass ratio is approximately 1:2 and Si:O molar ratio is 1:3.

**7.1.** Calculate the empirical formula of beryl.

Hydroxyapatite is the mineral containing the largest amount of calcium and phosphorous inside the human body; it is part of the bonds and the dental enamel. It is formed by hydroxide phosphate and calcium ions. Its composition in mass percent is 39,9% of calcium and 18,5% phosphorous.

**7.2.** Calculate the empirical formula of hydroxyapatite.

Borax is used to manufacture of enamel, glass and ceramic; it can be easily transformed into boric acid, which has numerous uses. Borax contains the chemical elements boro, sodium, hydrogen and oxygen; and can be considered a hydrated mix sodium and boron oxide. 1,91 g of borax react with 0,0100 mol of hydrochloric acid, producing 1,25 g of boric acid.

**7.3.** Calculate the empirical formula of borax.  $M(\text{H}_3\text{BO}_3) = 61,84 \text{ g} \cdot \text{mol}^{-1}$